# Experimental Analysis of Public Policies

## Session 1

Foundations of Policy Evaluation

Etienne Dagorn
University of Lille

# Session Objectives

By the end of this session, students should be able to:

- translate a substantive policy question into unit, treatment, outcome, population, horizon, and decision margin;
- define ATE, ATT, and ATU in words and notation, with ITT;
- use potential outcomes to state the missing counterfactual precisely;
- explain when a treatment effect is not well defined because of vague treatment versions, timing, or interference;
- decompose a naive comparison into a causal component and selection bias;
- explain why random assignment is the benchmark for internal validity and why external validity is a separate question.

# Roadmap

From Policy Questions to Causal Estimands

Potential Outcomes and the Missing Counterfactual

When Is a Treatment Effect Well Defined?

Selection, Identification, and the Experimental Benchmark

From Experiments to Real-World Policy Evaluation

# Roadmap

From Policy Questions to Causal Estimands

Potential Outcomes and the Missing Counterfactual

When Is a Treatment Effect Well Defined?

Selection, Identification, and the Experimental Benchmark

From Experiments to Real-World Policy Evaluation

# Headline to Estimand

**Two minutes**

The mayor says: "Free tutoring raised scores, so we should scale it."

1. What is the unit?
2. Is the policy lever an offer, receipt, or hours?
3. What outcome and horizon matter?
4. Which pupils define the target population?

# Stage 1: A City Introduces Tutoring

- The city funds free after-school tutoring for eligible pupils.
- The mayor asks: "Does tutoring work?"
- The education department has test scores, school records, and attendance logs.

**First design reaction**

"Does tutoring work?" is a policy motivation, not yet a causal estimand.

**Why:** the answer changes if we mean an offer, actual attendance, 30 hours of tutoring, short-run scores, graduation, participants, or all eligible pupils.

# The Policy-Evaluation Objects

| Object | Tutoring example |
|---|---|
| Unit of analysis | pupil, classroom, school, or cohort |
| Treatment | tutoring offer, participation, or hours attended |
| Outcome | test score, grades, attendance, graduation, later earnings |
| Target population | eligible pupils in the city, in pilot schools, or nationally |
| Time horizon | end of semester, one year, or longer term |
| Policy decision | continue, scale, retarget, redesign, or stop |
| Causal estimand | the average effect that matches the decision |

# Changing One Object Changes the Question

| Change | Different evaluation question |
|---|---|
| Outcome: test score vs graduation | learning this year or educational trajectory? |
| Population: pilot schools vs all eligible pupils | programme performance or scale-up? |
| Horizon: semester vs two years | immediate gains or persistent gains? |
| Treatment: offer vs attendance | policy access or treatment receipt? |
| Unit: pupil vs school | individual tutoring or school-level programme? |

**Key point**

A causal effect is always attached to a defined intervention, population, outcome, and time.

# Treatment Is Not One Word

$$Z_i = \text{being offered tutoring}$$

$$D_i = \text{actually participating in tutoring}$$

$$H_i = \text{number of tutoring hours attended}$$

- $Z_i$ is the offer or assignment that the policy can control.
- $D_i$ is treatment received.
- Each can answer a legitimate but different policy question.
- A design is ambiguous if it calls all three simply "treatment."

# From Question to Estimand

**Working definition**

A causal estimand is the causal quantity we want to learn before looking at the data.

- It is not the regression coefficient by default.
- It is chosen by the policy decision and the population of interest.
- It must specify the comparison between potential outcomes.

**Example:** among all eligible pupils, what is the average effect of being offered tutoring this school year on end-of-year maths scores?

# Roadmap

From Policy Questions to Causal Estimands

Potential Outcomes and the Missing Counterfactual

When Is a Treatment Effect Well Defined?

Selection, Identification, and the Experimental Benchmark

From Experiments to Real-World Policy Evaluation

# Potential Outcomes

$$Y_i(1) = \text{outcome for pupil } i \text{ if treated}$$

$$Y_i(0) = \text{outcome for pupil } i \text{ if not treated}$$

- Each pupil has two potential outcomes for the same outcome and horizon.
- The notation forces us to name the counterfactual state.
- The treatment definition must already be clear before we write $Y_i(1)$ and $Y_i(0)$.

# Individual Treatment Effects

$$\tau_i = Y_i(1) - Y_i(0)$$

- $\tau_i$ is the individual causal effect.
- It compares two potential outcomes for the same pupil.
- It is conceptually simple and empirically impossible to observe directly.

**The problem**

For the same pupil at the same time, we never observe both $Y_i(1)$ and $Y_i(0)$.

# Observed Outcome

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

**If treated**

$$Y_i = Y_i(1)$$

The untreated outcome $Y_i(0)$ is missing.

**If untreated**

$$Y_i = Y_i(0)$$

The treated outcome $Y_i(1)$ is missing.

# The Missing Counterfactual in a Table

| Pupil | $Y_i(1)$ | $Y_i(0)$ | $\tau_i$ | $D_i$ | $Y_i$ |
|---|---|---|---|---|---|
| Alice | 78 | | | 1 | 78 |
| Benoit | | 62 | | 0 | 62 |
| Chloe | 84 | | | 1 | 84 |
| David | | 71 | | 0 | 71 |

**Read the table**

Observed outcomes are real data.
The missing cells are counterfactuals.
Individual effects require both cells, so we estimate average effects using design.

# The Missing Counterfactual in a Table

| Pupil | $Y_i(1)$ | $Y_i(0)$ | $\tau_i$ | $D_i$ | $Y_i$ |
|---|---|---|---|---|---|
| Alice | 78 | ? | | 1 | 78 |
| Benoit | ? | 62 | | 0 | 62 |
| Chloe | 84 | ? | | 1 | 84 |
| David | ? | 71 | | 0 | 71 |

**Read the table**

Observed outcomes are real data.
The missing cells are counterfactuals.
Individual effects require both cells, so we estimate average effects using design.

# The Missing Counterfactual in a Table

| Pupil | $Y_i(1)$ | $Y_i(0)$ | $\tau_i$ | $D_i$ | $Y_i$ |
|---|---|---|---|---|---|
| Alice | 78 | ? | ? | 1 | 78 |
| Benoit | ? | 62 | ? | 0 | 62 |
| Chloe | 84 | ? | ? | 1 | 84 |
| David | ? | 71 | ? | 0 | 71 |

**Read the table**

Observed outcomes are real data.
The missing cells are counterfactuals.
Individual effects require both cells, so we estimate average effects using design.

# Average Treatment Effect

$$ATE = E[Y_i(1) - Y_i(0)]$$

- The ATE averages individual effects over the target population.
- It answers a scale-up style question: what would happen, on average, if eligible pupils were treated rather than untreated?
- The target population must be named for the ATE to be interpretable.

**Tutoring example:** the average effect of tutoring for all pupils eligible under the city's rule.

# ATT and ATU

$$ATT = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$
$$ATU = E[Y_i(1) - Y_i(0) \mid D_i = 0]$$

- ATT: effect for pupils who actually receive treatment.
- ATU: effect untreated pupils would have if treated.
- They can differ when treatment effects are heterogeneous or selection is systematic.

# Estimands as Policy Questions

| Policy question | Relevant estimand |
|---|---|
| What is the effect on current participants? | ATT |
| What would be the effect if eligible pupils were treated? | ATE |
| What would happen to current non-participants if they took up tutoring? | ATU |

**Choice rule**

The estimand should match the decision margin: evaluate current performance, expand coverage, target non-participants, or change eligibility.

# Interactive: Which Estimand?

1. The city asks whether tutoring helped pupils who enrolled.
2. The city asks whether every eligible pupil should be guaranteed a place.
3. The city asks whether non-participants should receive stronger encouragement.

# Interactive: Which Estimand? Debrief

**Answers**

1. ATT.
2. ATE of the offer/access policy; under full compliance, this coincides with a treatment-receipt ATE.
3. ATU or an offer effect, depending on the policy lever.

# Heterogeneous Effects

$$\tau_i = Y_i(1) - Y_i(0) \quad \text{may differ across pupils.}$$

| Group | Untreated risk | Effect |
|---|---|---|
| High baseline score | low | small |
| Middle baseline score | medium | moderate |
| Low baseline score | high | large |

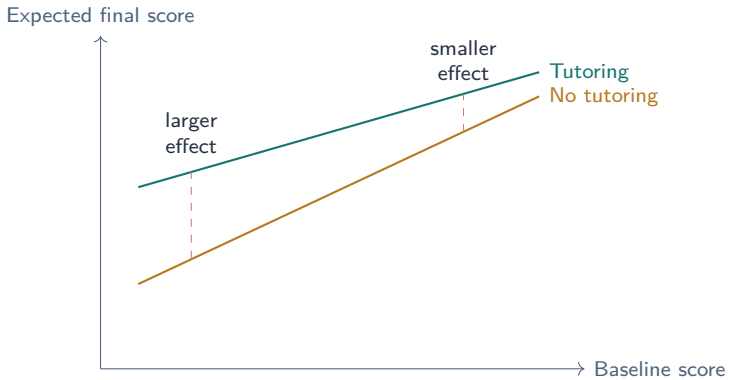
**Preview**

Average effects can differ across populations because individual effects can differ across pupils.

- $X_i$ is a pre-treatment characteristic.
- Effects in one group need not equal effects in another.

**Implication:** ATE, ATT, ATU, and effects in another population may differ.

# A Simple Interaction Picture



# Roadmap

From Policy Questions to Causal Estimands

Potential Outcomes and the Missing Counterfactual

When Is a Treatment Effect Well Defined?

Selection, Identification, and the Experimental Benchmark

From Experiments to Real-World Policy Evaluation

# Before Identification: Define the Treatment

**Conceptual order**

First ask whether the causal effect is well defined. Then ask whether it is identified.

- What exactly happens to a treated pupil?
- Can untreated pupils receive something similar?
- Can one pupil's treatment change another pupil's outcome?
- Are treatment, covariates, and outcomes ordered correctly in time?

# Consistency

**Consistency**

If pupil $i$ actually receives treatment level $d$, then the observed outcome equals the potential outcome under that level:

$$Y_i = Y_i(d).$$

- This sounds obvious only when $d$ is well defined.
- "Tutoring" must correspond to a clear intervention.
- Otherwise $Y_i(1)$ bundles together many different versions of treatment.

# Multiple Versions of Treatment

| Version called “tutoring” | Why it may matter |
|---|---|
| 1 hour online each month | low intensity, limited personalization |
| 30 hours face-to-face | much higher exposure |
| Small-group tutoring | peer interaction and lower cost |
| One-to-one tutoring | more individualized attention |
| Exam-preparation tutoring | short-run test strategy vs learning |

**Interpretation**

If different versions have different effects, the estimate describes the implemented package, not an abstract word.

# No Interference

**Standard notation**

$$Y_i(d_i)$$

Pupil $i$'s outcome depends only on pupil $i$'s treatment.

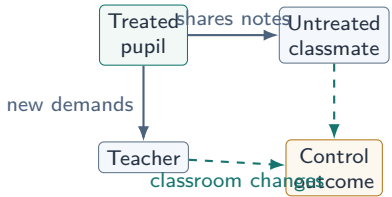
**More general world**

$$Y_i(d_1, \ldots, d_N)$$

Pupil $i$'s outcome may depend on classmates' treatment too.

- Treated pupils may share materials with untreated classmates.
- Teachers may change classroom instruction because some pupils receive tutoring.
- Programme capacity may change school resources.

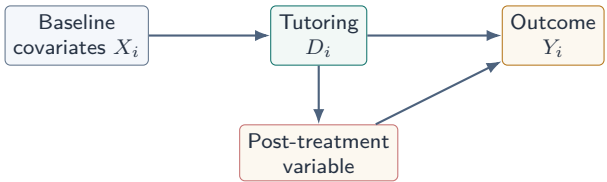
# Classroom Spillovers




**Design question**

Should the evaluation randomize pupils, classrooms, or schools if peer spillovers are likely?

# Timing




- Specify when treatment starts.
- Specify when the outcome is measured.
- Check that control variables are truly pre-treatment.

# Well-Defined Effect Checklist

1. Is the treatment version specific enough to define $Y_i(1)$?
2. Is the comparison condition specific enough to define $Y_i(0)$?
3. Is the unit of analysis clear?
4. Are outcomes measured after treatment?
5. Could one unit's treatment affect another unit's outcome?
6. Does the estimand match the policy implementation?

**Stage 2**

Before debating selection bias, revise the tutoring protocol so another team could implement the same treatment.

# Roadmap

From Policy Questions to Causal Estimands

Potential Outcomes and the Missing Counterfactual

When Is a Treatment Effect Well Defined?

Selection, Identification, and the Experimental Benchmark

From Experiments to Real-World Policy Evaluation

# Stage 2: Participation Is Voluntary

- The city opens tutoring places to eligible pupils.
- Some families enrol; others do not.
- Schools promote the programme with different intensity.

**Tempting comparison**

Compare participants and non-participants at the end of the year.

**Design question:** do non-participants represent what participants would have scored without tutoring?

# The Naive Comparison

$$\Delta^{naive} = E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$$

- Easy to compute.
- Useful as a descriptive starting point.
- Causal only if untreated potential outcomes are comparable across groups.

**Ask**

Before seeing any estimate, should we expect participants to have higher or lower $Y_i(0)$ than non-participants?

# Selection-Bias Decomposition

$$
\begin{aligned}
E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] &= E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] \\
&= E[Y_i(1) - Y_i(0) \mid D_i = 1] \\
&\quad + E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0].
\end{aligned}
$$

**ATT**

$$E[Y_i(1) - Y_i(0) \mid D_i = 1]$$

**Selection bias**

$$E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

# What the Decomposition Teaches

- The naive comparison is not “wrong” mathematically.
- It answers a descriptive question unless the selection-bias term is zero.
- Selection bias is about untreated potential outcomes, $Y_i(0)$.
- Its sign depends on how pupils enter or are targeted into tutoring.

**Identification**

An identification argument explains why an observed comparison equals the intended causal estimand under explicit assumptions.

# Interactive Sign of Bias: Scenario A

**Scenario A**

Highly motivated families are more likely to enrol in tutoring.

**Question:**

$$E[Y_i(0) \mid D_i = 1] \quad \text{is greater or smaller than} \quad E[Y_i(0) \mid D_i = 0]?$$

**Answer**

**Implication**

# Interactive Sign of Bias: Scenario A

**Scenario A**

Highly motivated families are more likely to enrol in tutoring.

**Question:**

$$E[Y_i(0) \mid D_i = 1] \quad \text{is greater or smaller than} \quad E[Y_i(0) \mid D_i = 0]?$$

**Answer**

Likely greater: participants would have scored higher even without tutoring.

**Implication**

# Interactive Sign of Bias: Scenario A

**Scenario A**

Highly motivated families are more likely to enrol in tutoring.

**Question:**

$$E[Y_i(0) \mid D_i = 1] \quad \text{is greater or smaller than} \quad E[Y_i(0) \mid D_i = 0]?$$

**Answer**

Likely greater: participants would have scored higher even without tutoring.

**Implication**

The naive participant–non-participant comparison may overstate the effect.

# Interactive Sign of Bias: Scenario B

**Scenario B**

Schools target tutoring at pupils who are already struggling.

**Question:**

$$E[Y_i(0) \mid D_i = 1] \quad \text{is greater or smaller than} \quad E[Y_i(0) \mid D_i = 0]?$$

**Answer**

**Implication**

# Interactive Sign of Bias: Scenario B

**Scenario B**

Schools target tutoring at pupils who are already struggling.

**Question:**

$$E[Y_i(0) \mid D_i = 1] \quad \text{is greater or smaller than} \quad E[Y_i(0) \mid D_i = 0]?$$

**Answer**

Likely smaller: treated pupils would have scored lower without tutoring.

**Implication**

# Interactive Sign of Bias: Scenario B

**Scenario B**

Schools target tutoring at pupils who are already struggling.

**Question:**

$$E[Y_i(0) \mid D_i = 1] \quad \text{is greater or smaller than} \quad E[Y_i(0) \mid D_i = 0]?$$

**Answer**

Likely smaller: treated pupils would have scored lower without tutoring.

**Implication**

The naive comparison may understate the effect.

# Stage 3: Capacity Is Limited

- Eligible demand exceeds available tutoring places.
- The city randomly assigns offers among eligible pupils.
- Random assignment is decided before outcomes are realised.
- For this clean benchmark, suppose every offered pupil attends: $D_i = Z_i$.

**Experimental benchmark**

$$Z_i \perp (Y_i(1), Y_i(0))$$

Assignment is independent of the potential outcomes relevant to the treatment contrast.

**Interpretation:** assigned arms are comparable in expectation because the offer does not use expected outcomes.

# Why Random Assignment Works

If

$$Z_i \perp (Y_i(1), Y_i(0)),$$

then

$$E[Y_i(0) \mid Z_i = 1] = E[Y_i(0) \mid Z_i = 0]$$

and

$$E[Y_i(1) \mid Z_i = 1] = E[Y_i(1) \mid Z_i = 0].$$

- The selection-bias term is zero in expectation.
- Treatment and control groups can stand in for each other's missing potential outcomes.
- In the full-compliance benchmark, $D_i = Z_i$, so assigned arms are also receipt groups.

# What the Difference in Means Identifies

$$E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] = E[Y_i(1) - Y_i(0)]$$

- With full compliance, no interference, and observed outcomes, assigned-arm means identify the average treatment-receipt effect in the randomised study sample.
- For a fixed experimental sample, the sample difference estimates SATE in expectation over the randomisation.
- If the randomised sample represents a larger target population, it can also inform PATE.
- Randomisation solves baseline selection; it does not by itself solve attrition, spillovers, or vague treatments.

# Roadmap

From Policy Questions to Causal Estimands

Potential Outcomes and the Missing Counterfactual

When Is a Treatment Effect Well Defined?

Selection, Identification, and the Experimental Benchmark

From Experiments to Real-World Policy Evaluation

# Stage 4: Offers Are Randomised, Attendance Is Not

$Z_i$ = randomised offer or assignment $\qquad$ $D_i$ = actual participation

- The city controls who receives an offer.
- Pupils and families decide whether to attend.
- Suppose only 60% of pupils offered tutoring attend.

**Question**

What effect does the experiment identify immediately: the effect of the offer or the effect of attending?

# Preview: Intention-to-Treat

$$\text{Under randomised } Z: \quad ITT = E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]$$

- Here potential outcomes are indexed by the offer $Z$, not by receipt $D$.
- The causal ITT answers: what is the effect of being offered the policy?
- The observed assigned-arm contrast identifies it under random assignment, no interference, and observed outcomes.
- It includes take-up behaviour and implementation frictions.

# Stage 5: Pupils Share Materials

- Treated pupils share worksheets with untreated classmates.
- Teachers adjust classroom examples after seeing tutoring content.
- Control pupils may no longer represent "no exposure to tutoring."

What does the treatment-control difference estimate now?

**Lesson**

Real-world policy evaluation depends on the treatment environment, not only the assignment rule.

# Stage 6: Some Outcomes Are Missing

**Scenario**

Twenty percent of pupils do not take the final test.

- Random assignment solved baseline treatment selection.
- Missing outcomes can create a new selection problem after assignment.
- The threat is sharper if attrition differs by assignment or expected outcome.

Is random assignment enough if outcomes are missing selectively?

# What Randomisation Does Not Automatically Solve

| Issue | Why it matters |
|---|---|
| Non-compliance | assignment differs from treatment receipt |
| Attrition | observed outcomes may be selected after assignment |
| Spillovers | control outcomes may be affected by treated units |
| Multiple treatment versions | $Y_i(1)$ may mix different interventions |
| Measurement | outcome changes may reflect measurement changes |
| External validity | the effect may not travel to another context |

# Random Assignment vs Random Sampling

**Random assignment**

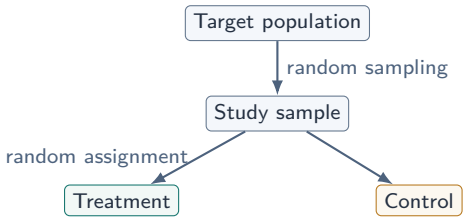
Who receives treatment?

**Purpose:** internal validity and causal identification.

**Random sampling**

Who enters the study?

**Purpose:** representativeness and generalisation.



# Preview: Four Possible Designs

| Design | What it can support |
|---|---|
| Random assignment, no random sampling | strong internal validity, narrower generalisation |
| Random sampling, no random assignment | representative description, weak causal identification |
| Both random sampling and assignment | strong design for causality and population inference |
| Neither | requires a careful observational identification argument |

**Common confusion**

Random sampling is not required for internal validity. Random assignment is not enough for external validity.

# Internal and External Validity in This Framework

| Validity question | Framework translation |
|---|---|
| Internal validity | Does the design identify the intended causal effect in the study? |
| External validity | To what population, setting, implementation, or policy context does the effect generalise? |

- External validity is broader than sample representativeness.
- It also depends on treatment-effect heterogeneity, implementation quality, institutional context, and target population.
- A strong experiment can answer a narrow policy question very well.

# Stage 7: Should the City Scale Nationally?

- The pilot used trained tutors in motivated schools.
- National rollout may involve different pupils, schools, budgets, and tutors.
- The treatment version and target population may change.

> **External-validity question**
>
> Is the pilot effect informative for the national policy environment, or only for the pilot as implemented?

**Policy interpretation:** scaling a programme changes who receives it, who delivers it, and sometimes what the treatment actually is.

# Evolving Tutoring Case: Full Arc

| Stage | Policy situation | Design question |
|---|---|---|
| 1 | City introduces tutoring | What is the estimand? |
| 2 | Participation is voluntary | Why is selection a threat? |
| 3 | Offers are randomised | What does randomisation solve? |
| 4 | Only 60% attend | Is treatment offer or receipt? |
| 5 | Pupils share materials | What assumption may fail? |
| 6 | 20% miss the final test | What new selection threat appears? |
| 7 | National expansion is considered | Where does the estimate apply? |

**Teaching point**

The same policy generates different causal questions as implementation details become visible.

# How to Read a Policy-Evaluation Paper

1. What policy decision motivates the study?
2. What exactly are $Z$, $D$, $Y$, $X$, the population, and the horizon?
3. What causal estimand is being targeted?
4. What is the missing counterfactual?
5. What creates identifying variation?
6. What assumption makes the comparison causal?
7. What diagnostics or threats should change our confidence?
8. To whom and where does the estimate apply?

**Reading habit**

Do not start by asking whether the table is significant. Start by asking what causal object the table is trying to estimate.

# Final Design Logic

| | |
|---|---|
| **Research question** | What exactly are we trying to learn when evaluating a policy? |
| **Ideal experiment** | Randomly assign a clearly defined offer or treatment within the study population. |
| **Counterfactual** | The same units under the other treatment state. |
| **Source of variation** | The random assignment, or a comparison justified by a later design. |
| **Identification assumption** | The comparison units are exchangeable in potential outcomes for the intended estimand. |
| **Main threat** | Vague treatment versions, spillovers, non-compliance, attrition, measurement, and limited external validity. |

# Transition to Session 2

- We now know what causal effect we want.
- We know what the ideal experiment would look like.
- We know why selection creates bias in voluntary participation.

> **Next**
>
> What can we do when treatment is not randomised but we observe rich pre-treatment covariates?

**Bridge:** regression adjustment and matching will try to build a credible comparison group using observed baseline information.